

rad21 (SCHPO, P30776) — Review of GO:1990342 "heterochromatin island" IDA annotation

Focus: existing_go_annotation_decision · hypothesis slug `existing-go-1990342-undecided`

Annotation under review (PomBase detail 472247): rad21 (SPCC338.17c/P30776) *is_active_in heterochromatin island* (GO:1990342), evidence IDA (ECO:0000314), reference

P 22144463

extension *exists_during* meiotic cell cycle (GO:0098762), assigned_by PomBase 2020-11-26, low throughput. **Seed hypothesis:** the annotation should receive review action **UNDECIDED**.

Executive Judgment

Verdict: SUPPORTED. The annotation is directly confirmed by the primary paper's own ChIP data. UNDECIDED is no longer warranted — the recommended action is ACCEPT (with one minor caveat on the temporal extension).

The central knowledge gap — whether

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actually contains a Rad21 localization assay, which was inaccessible from the abstract — was **resolved by reading the full text** (obtained via PMC6338074). The paper states verbatim:

"Factors that bind H3K9me and their associated effectors localize to heterochromatin islands: Chromatin immunoprecipitation (ChIP) detected the Clr4 complex (ClrC) subunit Raf2, Swi6/HP1, as well as ... RITS (Chp1 and Ago1) and SHREC (Clr3 and Mit1) ... at heterochromatin islands (fig. S2). ... **We also found the cohesin-loading factor Mis4, which interacts with Swi6/HP1 (10), and the cohesin sub unit Rad21 enriched at meiotic heterochromatin islands (fig. S2).**"

This is an explicit ChIP-based (IDA-appropriate) demonstration that Rad21 — and its loader Mis4 — are enriched at heterochromatin islands. It fully validates the PomBase IDA annotation and explains the internal curation pattern found in Iteration 2 (both the cohesin kleisin **and** its loader annotated to islands from this single reference). The abstract's silence on Rad21 (the entire basis for the seed's caution) is simply because the cohesin result is a supplementary-figure finding, not an abstract-level headline.

Caveats / remaining minor issue: - The ChIP was performed in **log-growing (vegetative) cells**; the islands *cover meiotic-gene loci* ("meiotic heterochromatin islands") but the localization is a vegetative-growth phenomenon. The rad21-specific **exists_during meiotic cell cycle (GO:0098762) extension** is therefore likely a minor mis-extension (Rad21 is also largely replaced by Rec8 during actual meiosis; Kitajima 2003, Yokobayashi 2003). The **base CC term is fully supported**; the temporal qualifier should be reviewed/removed independently. - Confidence in the base term is now **high/direct**; confidence that the extension is correct is **low**.

Evidence Matrix

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence/limitations
P 22144463 (Zofall 2012, Science), full text Fig. S2	Direct assay (ChIP)	SUPPORTS (decisive)	rad21 localizes to heterochromatin islands	"the cohesin subunit Rad21 enriched at meiotic heterochromatin islands (fig. S2)"; Mis4 loader also enriched	S. pombe, vegetative (log-growing) cells	Direct evidence covering genes is vegetative temporal extension question
PomBase term GO:1990342 (API)	Database/curation pattern	Supports	rad21 IDA basis is genuine	P 22144463 incl. both Rad21 (kleisin) and Mis4 (loader) + Swi6/Chp1	S. pombe (PomBase 2020)	Correctly directed
GEO GSE33404 (NCBI eutils)	Dataset audit	Qualifies	Genomic data underlying the paper	Only 5 H3K9me2 array samples; Rad21/Mis4 shown by targeted ChIP (fig. S2) , not arrays	S. pombe	Explains cohesin figure
P 11780129 (Nonaka 2002, Nat Cell Biol)	Interaction + localization	Supports (mechanism)	Heterochromatin recruits cohesin	Swi6/HP1 recruits cohesin; Psc3–Swi6 interaction	S. pombe centromere/mat	Mechanism basis at Swi6 island ref 10 paper
P 21189291 (Dheur 2011, MCB)	Mutant phenotype + localization	Supports (general)	Rad21 acts in facultative-type heterochromatin	rad21 mutation depletes Swi6/H3K9me across a subtelomeric domain	S. pombe subtelomere	Different class island

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence/limitations
P 31278118 (Folco 2019, Genetics)	Localization (ChIP) + genetics	Supports (general)/Competing	Rad21 within heterochromatin	Genome-wide Rad21/Psc3 within heterochromatin; can <i>impede</i> assembly in pds5Δ	S. pombe	Consistent with heterochromatin role in gene silencing; dependent on Psc3
P 25307058 (Mizuguchi 2014, Nature)	Hi-C + ChIP	Supports (general)	Cohesin at heterochromatin	Cohesin loaded at pericentromeric/subtelomeric heterochromatin	S. pombe	Not inconsistent with specific role in chromosome architecture
P 12750522 (Kitajima 2003, Nature)	Localization/genetics	Refutes/conflicts (extension only)	rad21 active during meiotic cell cycle	Rec8 largely replaces Rad21/Scc1 in meiosis	S. pombe meiosis	Conflicts with extension of the CEN-kinetochore
P 12748297 (Yokobayashi 2003, Dev Cell)	Localization/genetics	Qualifies (extension)	rad21 in meiosis	Rad21 substitutes only at centromeres when Rec8 absent	S. pombe meiosis	Meiotic role is substituted; limited to centromeres

(Provenance: PMC6338074 full-text retrieval with verbatim quote; PomBase REST API annotation + term records; NCBI eutils GDS esummary for GSE33404. CSVs: `evidence_matrix_rad21_G01990342.csv`, `go_decision_table_rad21.csv`, `island_genes_from_PMID22144463.csv`.)

GO Curation Implications

This is a **cellular component (CC)** annotation. Lead (requires curator verification):

- **ACCEPT / RETAIN GO:1990342** for rad21. The IDA is directly supported by ChIP data in Fig. S2 of

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Change the review action from **UNDECIDED** → **ACCEPT**.

- **Review the `exists_during` meiotic cell cycle (GO:0098762) extension** (present only on the rad21 annotation). The assay is in vegetative cells and islands merely *cover* meiotic genes; consider **removing or replacing** this temporal qualifier (e.g., no temporal extension, or a `coincident_with` meiotic-gene locus extension as used for other island genes in the same term).
 - This is not a "protein binding" case — the informative content is the CC localization term, which is now well-supported.
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Mechanistic Scope

- **Direct molecular event (confirmed):** the cohesin kleisin Rad21 (with loader Mis4) is physically enriched by ChIP at facultative heterochromatin islands — small H3K9me/Swi6 domains over meiotic genes silenced during vegetative growth.
 - **Upstream vs. downstream:** island *formation* is driven by RNA elimination (Mmi1/Red1) + Clr4/H3K9me; Swi6/HP1 then binds H3K9me and recruits downstream effectors, **including cohesin (Rad21/Mis4)**. Rad21 is thus a *recruited effector at* islands, not an island-forming factor — but the CC term "is_active_in heterochromatin island" is exactly a localization statement and is correct.
 - The mechanistic chain (H3K9me → Swi6 → Psc3/Mis4–Swi6 interaction → cohesin) is consistent with Nonaka 2002 (cited as ref 10 in the paper).
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Conflicts and Alternatives

- **Temporal-extension conflict (only remaining concern):** `exists_during` meiotic cell cycle conflicts with (i) the vegetative-cell context of the actual ChIP and (ii) Rad21→Rec8 replacement in meiosis. The base localization is unaffected.
 - **No paralog confusion for the base term:** the paper explicitly assays Rad21 (mitotic kleisin), not Rec8; the vegetative context is where Rad21, not Rec8, is the active kleisin.
 - **Context-dependent cohesin role:** Folco 2019 shows cohesin can antagonize heterochromatin elsewhere, but that does not affect the *localization* claim at islands.
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Knowledge Gaps

1. **Quantitative magnitude / genome-wide extent of Rad21 island enrichment.** Checked: the paper shows targeted ChIP (fig. S2); GSE33404 has no cohesin arrays. Matters only for strength, not existence, of the annotation. *Resolve by:* intersecting published Rad21/Psc3 ChIP-seq (Mizuguchi 2014; Folco 2019) with island coordinates.
 2. **Intended meaning/validity of the `exists_during` meiotic cell cycle extension.** Checked: assay is vegetative; conflicts with Rec8 replacement. *Resolve by:* curator review of the extension against the figure legend and Materials.
 3. **Functional role of cohesin at islands (vs. mere presence).** Checked: not addressed in this paper. Not required for a CC localization annotation. *Resolve by:* rad21-ts/mis4-ts effects on island H3K9me and meiotic-gene silencing.
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Discriminating Tests

- **(Already done here) Full-text/Fig. S2 inspection** — decisive; confirms Rad21 at islands.
- **Coordinate-overlap analysis:** intersect published Rad21/Psc3 ChIP-seq peaks with the ~21 island coordinates; permutation test for enrichment (quantifies the effect genome-wide).
- **Perturbation:** test island Rad21 enrichment in swi6Δ/clr4Δ (predicted loss, per Nonaka 2002 logic) to confirm the recruitment mechanism.

- **Rec8 vs. Rad21 discrimination** only needed if the meiotic-timing extension is to be retained.

Curation Leads (require curator verification)

- **Action change:** UNDECIDED → ACCEPT for *rad21 is_active_in* heterochromatin island (GO:1990342).
- **Confirming reference + exact snippet to verify:**

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Fig. S2 / Results — *"the cohesin sub unit Rad21 enriched at meiotic heterochromatin islands (fig. S2)"* and *"We also found the cohesin-loading factor Mis4, which interacts with Swi6/HP1 (10) ... enriched at meiotic heterochromatin islands."* (Text obtained from PMC6338074; curator should confirm against the published Fig. S2.)

- **Supporting mechanistic reference:**

P 11780129

— Swi6/HP1 recruits cohesin to heterochromatin (paper's ref 10).

- **Separate flag:** review/remove the `exists_during` meiotic cell cycle (GO:0098762) extension on this annotation (assay is vegetative; Rad21→Rec8 in meiosis).
 - **Suggested question to annotator (ORCID 0000-0003-4148-4606):** confirm Fig. S2 as the IDA basis and clarify the intent of the meiotic-cell-cycle extension.
 - **Suggested experiment (optional, for functional depth):** *rad21/mis4* loss-of-function effect on island H3K9me and meiotic-gene silencing.
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Resolution note: Iterations 1–2 flagged this as unresolved/partially supported because the full text was believed inaccessible (Science 403; Europe PMC OA XML 404). Iteration 3 successfully retrieved the author-manuscript full text from PMC6338074, which contains the direct ChIP statement, converting the verdict to SUPPORTED. The only residual issue is the temporal extension.
